■ ParkIQ

AI-Powered Parking Enforcement Intelligence

Gridlock Hackathon 2.0 Round 2 | Problem Statement 1: Parking-Induced Congestion

298,450	8,173	54	6
BTP Violation Records	ASTRAM Incidents	Police Stations	Dashboard Tabs

The Problem

Bengaluru loses 60+ minutes/day per commuter to traffic congestion

■ No Visibility	BTP has no real-time map of illegal parking hotspots — officers operate blind.
■ Reactive Deployment	Officers sent on complaints, not data — high-impact zones go unpatrolled.
■ No Impact Measurement	No metric linking parking violations to downstream congestion events.
■ Missed Peak Windows	Morning rush (9–11 AM) = highest violation density, but officer deployment doesn't match.

Solution: ParkIQ Dashboard

6-tab Streamlit platform — raw violation data to deployable patrol schedules

■■ Live Heatmap Violation density+severity, 3 view modes	**■ Hotspot Analysis** Priority-scored junctions, 4-factor model
■ Temporal Patterns Hourly/daily/monthly violation profiles	**■ Impact Quantification** ASTRAM cross-analysis + vehicle-hour loss
■ Enforcement Optimizer Shift-wise schedule for N officers	**■ Vehicle Intelligence** Type breakdown + cross-hour heatmap

Data Foundation

Two real BTP datasets — same 54 police stations, same time window

Attribute	BTP Violations (PS1)	ASTRAM Incidents
Records	298,450 raw / 115K approved	8,173 events
Period	Nov 2023 – Apr 2024	Nov 2023 – Apr 2024
Key Fields	GPS, vehicle type, violation type, junction, police station	Event type, cause, start/resolve time, priority, corridor
Police Stations	54 stations	54 stations (exact match)
Format	Parquet (21MB, compressed from 105MB CSV)	Parquet (1.3MB)

Key Findings from 115,400 Approved Records

Top Hotspot BTP051 — Safina Plaza 15,000+ violations; #1 priority enforcement zone	**Peak Window** 10:00 AM IST Morning rush = highest enforcement opportunity daily
Vehicle Type Cars = top offender Wider lane blockage = higher congestion impact per violation	**Concentration** Top 5 junctions ≈ 40% Of all named-junction violations — targeting 5 zones solves most of problem
Enforcement Effect r = –0.14 correlation Active enforcement stations show fewer ASTRAM traffic incidents	**Coverage** 54 stations complete Full Bengaluru spatial coverage — no enforcement blind spots

Priority Scoring Methodology

Score = 0.40 × Frequency + 0.30 × Peak-Hour + 0.20 × Incident Corr + 0.10 × Severity

40%	**Frequency Score** Total violations ÷ max violations — identifies chronic repeat hotspots
30%	**Peak-Hour Concentration** Violations 8–11 AM ÷ total — targets morning rush enforcement
20%	**Incident Correlation** ASTRAM incident rate at same police station — links parking to real congestion
10%	**Vehicle Severity Score** Avg severity weight (Truck=3× vs Scooter=0.5) — prioritizes high-impact blockages

Enforcement Optimizer

Officers → Zones → Shifts → Map — deployed in under 60 seconds

1	INPUT	Dispatcher sets available officer count (5–30 via slider)
2	RANK	System ranks all junctions by composite priority score, descending
3	ASSIGN	Officers assigned to top N zones across 3 shifts: Morning Rush / Late Morning / Night Drive
4	DEPLOY	Interactive Folium map: each officer shown at assigned location, colour-coded by shift
5	IMPACT	Projected violations addressable per deployment cycle shown in real-time

Impact & Next Steps

Quantified Impact	Next Steps (Phase 2)
✓ 115,400 approved violations geo-mapped, ranked, and actionable	→ Live BTP API integration for real-time violation feed
✓ Top 5 junctions → ~40% of hotspot violations addressed with minimal resource spend	→ Predictive ML model: forecast next-hour hotspot zones
✓ Peak enforcement window identified: 10 AM daily = maximum ROI per officer-hour	→ Mobile officer app with push notifications for zone assignments
✓ Est. 2,000+ vehicle-hours/week saved by data-driven patrol vs random deployment	→ Feedback loop: fine-rate vs incident-rate post-deployment measurement
✓ Architecture is real-time ready — plug in BTP live feed, no code changes needed	→ Expand to PS2 (incident detection) + PS3 (signal optimization)